1 Ideal Over-the-Air Computation Baseline

Before introducing the practical impairments inherent to wireless channels, such as Inter-Symbol Interference (ISI), multipath fading, and time synchronization errors, it is essential to establish a baseline model of Over-the-Air (OTA) computation under ideal conditions. The foundational premise of OTA computing is its ability to reduce the communication latency of data aggregation from $\mathcal{O}(K)$ to $\mathcal{O}(1)$ by allowing K devices to transmit simultaneously [3].

In this idealized scenario, the wireless medium is assumed to be perfectly synchronized, lacking frequency-selective fading, and corrupted solely by Additive White Gaussian Noise (AWGN). This chapter aggregates the mathematical formulation, algorithmic MATLAB modeling, block-level Simulink architecture, and GNU Radio flowgraphs that demonstrate this ideal, unimpaired baseline. Establishing this benchmark is critical for quantifying the exact degradation caused by physical channel constraints in subsequent chapters.

1.a Mathematical Formulation of the Ideal Channel

Consider a distributed wireless sensor network consisting of K End Device (ED)s transmitting local analog values to a central Fusion Center (FC). Let $x_k(t)$ denote the continuous-time, real-valued signal transmitted by the k-th device. For analog OTA computation, uncoded Pulse Amplitude Modulation (PAM) is widely considered the optimal transmission scheme, as it linearly maps the sensor readings directly to the amplitude of the transmitted waveform [2], [4].

In an ideal Multiple Access Channel (MAC), we assume perfect Channel State Information (CSI) at the transmitters, which allows for flawless pre-equalization of phase shifts and path loss attenuation. Consequently, the electromagnetic waves superimpose perfectly at the receiving antenna of the FC. The received baseband signal $y(t)$ is mathematically expressed as the linear sum of the transmitted signals plus noise:

$$y(t) = \sum_{k=1}^{K} x_k(t) + n(t) \tag{1}$$

where $n(t) \sim \mathcal{N}(0, \sigma^2)$ represents the AWGN with zero mean and variance σ^2. Because there is no multipath delay spread in this baseline model, the channel impulse response is effectively a Dirac delta function $h(t) = \delta(t)$. This implies that the current symbol is entirely unaffected by past or future symbols, resulting in zero ISI. To extract the arithmetic average of the K sensor readings, the FC simply applies a post-processing scaling factor of $1/K$ to the received aggregated signal $y(t)$.

1.b MATLAB Simulation and Code Analysis

To validate this superposition principle, the ideal environment can be modeled using MATLAB. The following script generates normally distributed simulated sensor readings, aligns them perfectly in time, and adds AWGN to emulate the ideal MAC.

Listing 1: MATLAB script for ideal OTA computation aggregation.

```

```

```
        % The Air acts as an ideal adder (MAC Superposition)
        Ideal_Sum = sum(X, 1);

        % Add AWGN
        Received_Signal = awgn(Ideal_Sum, SNR_dB, 'measured');

        % Calculate the Fusion Center Average
        Estimated_Average = Received_Signal / K;

        % Display Mean Squared Error (MSE)
        MSE = immse(Estimated_Average, mean(X, 1)); fprintf('Ideal Baseline MSE at %d dB SNR: %f\n', SNR_dB, MSE);
```

While Listing 1 demonstrates the fundamental principle in its unadulterated form, practical wireless systems necessitate pulse shaping to satisfy spectral constraints and facilitate signal recovery. Consequently, a more physically realistic model incorporates matched Raised Cosine (RC) filtering at both the transmitter and receiver. This architectural refinement introduces additional degrees of freedom—namely, the Symbols per Sample (sps) parameter and the roll-off factor (α)—which materially influence system performance. The enhanced implementation is presented in Listing 2.

Listing 2: MATLAB script for OTA computation aggregation using the RC filter.

```
        txUpSamped = upsample(X', sps_val); % phase may need to be set

        % Raised Cosine Filter
        rcFilter = rcosdesign(aleph, span, sps_val, "sqrt");

        rx_agg_signal = zeros(N * sps_val + length(rcFilter) - 1, 1);

        for k = 1:K
            rx_agg_signal = rx_agg_signal + conv(txUpSamped(:,k), rcFilter(:));
        end

        Received_Signal = awgn(rx_agg_signal, current_snr, 'measured');

        matchedSignal = conv(Received_Signal, flipud(rcFilter(:)), 'same');

        rxMatched = downsample(matchedSignal, sps_val);

        start_idx = floor(span/2); stop_idx = floor(span/2) + mod(span,2) + 1;

        % Truncate edges to avoid filter transient errors
        Estimated_Average = rxMatched(start_idx:end-stop_idx, :) / K;

        % Calculate and store MSE in the matrix
        mse_results(snr_idx, a_idx) = immse(Estimated_Average', mean(X, 1));
```

Computational execution of the aforementioned algorithms reveals a critical distinction between the idealized and practical regimes. When temporal synchronization is mathematically perfect and no filtering is applied, the estimation accuracy is entirely governed by the noise floor; the Mean Squared Error (MSE) exhibits a direct proportionality to the AWGN variance and inverse quadratic scaling with the number of transmitting sensors. Mathematically, this relationship is expressed as MSE $\propto \sigma^2/K^2$, which demonstrates the fundamental advantage of cooperative multi-device aggregation [1], [3]. Conversely, when pulse shaping filters are incorporated into the transmission chain, the interplay between symbol timing parameters (sps) and spectral efficiency (Roll-off Factor (α)) introduces measurable deviation from the ideal theoretical bound.

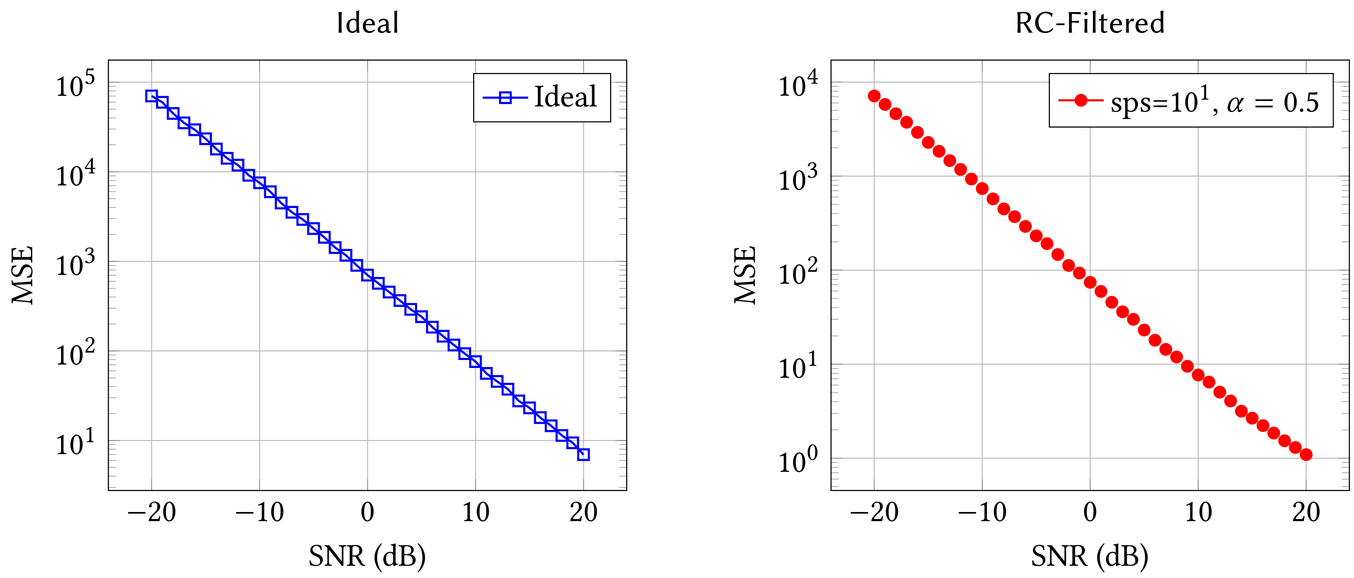

Figure 1.b presents the theoretical performance floor, wherein the MSE is a function solely of AWGN variance. In contrast, Figure 1.b illustrates the performance envelope when matched Raised Cosine filters are deployed at both transmitter and receiver, with sps and α held constant. Despite the introduction of filtering, the channel remains free of multipath distortion and timing misalignment, ensuring that noise contributions remain the dominant performance-limiting factor.

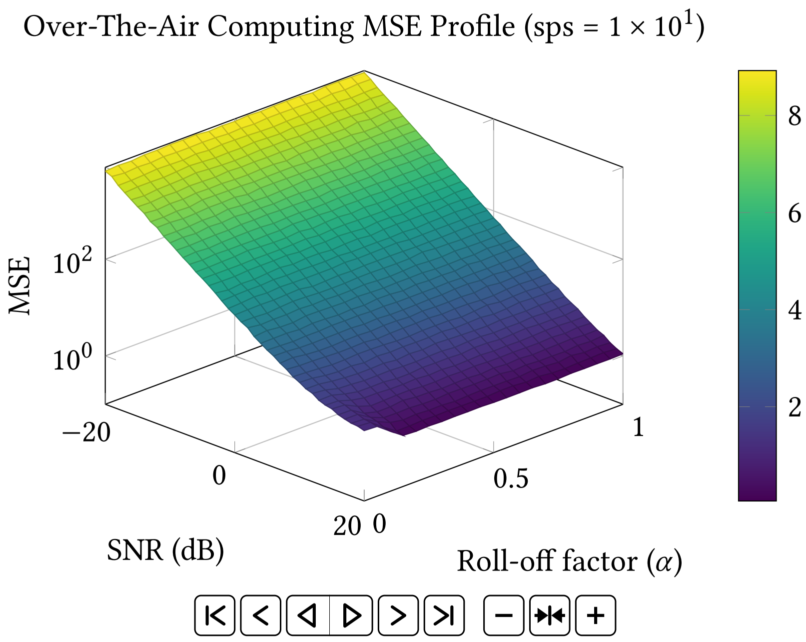

Figure 1: Interactive OTA computing animation demonstrating the architectural impact of Samples Per Symbol (sps) scaling.

A particularly noteworthy observation emerges from parametric analysis of the animation in Figure 1: while increasing sps generally yields marginal improvements in MSE performance across the SNR range examined, this enhancement asymptotically approaches a limit and does not constitute convergence toward the theoretical ideal baseline (absent pulse shaping). This behavior underscores a fundamental trade-off inherent to practical pulse-shaping implementations: enhanced spectral efficiency through reduced sps incurs amplified inter-symbol interference, whereas elevated oversampling ratios improve timing robustness at the cost of increased computational complexity and memory consumption.

1.c MATLAB Simulink Architecture

Transitioning from raw code to a block-level architecture, MATLAB Simulink provides a clear visual representation of the feedforward ideal system. In this model, the transmitters use standard PAM without the need for complex feedback loops or equalizers.

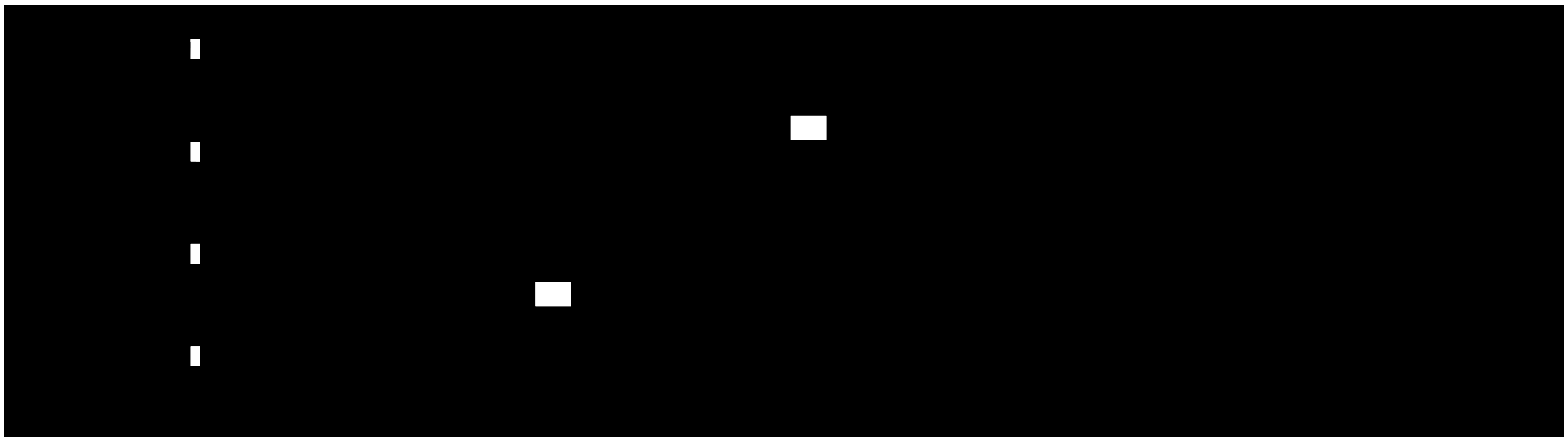

Figure 2: Simulink block diagram representing the ideal OTA computation network over an AWGN channel.

To bridge the gap between theory and practical system implementation, an enhanced model incorporating Raised Cosine pulse shaping is presented in Figure 3. This architecture employs matched filtering pairs at both transmission and reception stages, enabling quantitative assessment of performance degradation attributable to finite symbol-rate sampling and spectral roll-off constraints.

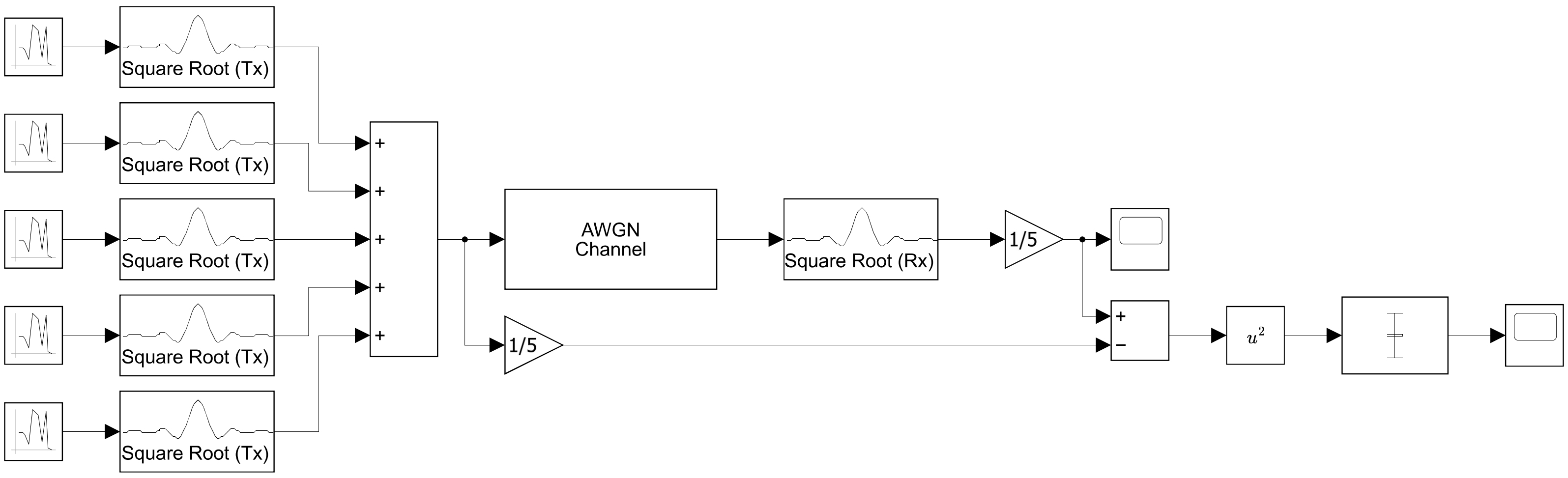

Figure 3: Simulink block diagram representing the ideal OTA computation network over an AWGN channel.

Inspection of the MSE calculation scope output demonstrates the quantitative deviation between the recovered aggregated estimate and the theoretical true value, reflecting the stochastic influence of the additive Gaussian noise. It is noteworthy that practical over-the-air computation mandates time-varying transmission (as opposed to static constant-valued signals), as pulse shaping through Finite Impulse Response (FIR) filtering inherently requires symbol-rate sampling and subsequent interpolation. Consequently, systems employing matched filter pairs necessitate carefully synchronized symbol generation to avoid spurious transient responses.

1.d GNU Radio Validation

Finally, the ideal model is verified using a Software Defined Radio (SDR) environment in GNU Radio. To maintain the ideal conditions mathematically defined in (1), the Root Raised Cosine (RRC) filters are perfectly matched. Specifically, the interpolation rate at the transmitters and the decimation rate at the receiver are identically matched to the symbol design

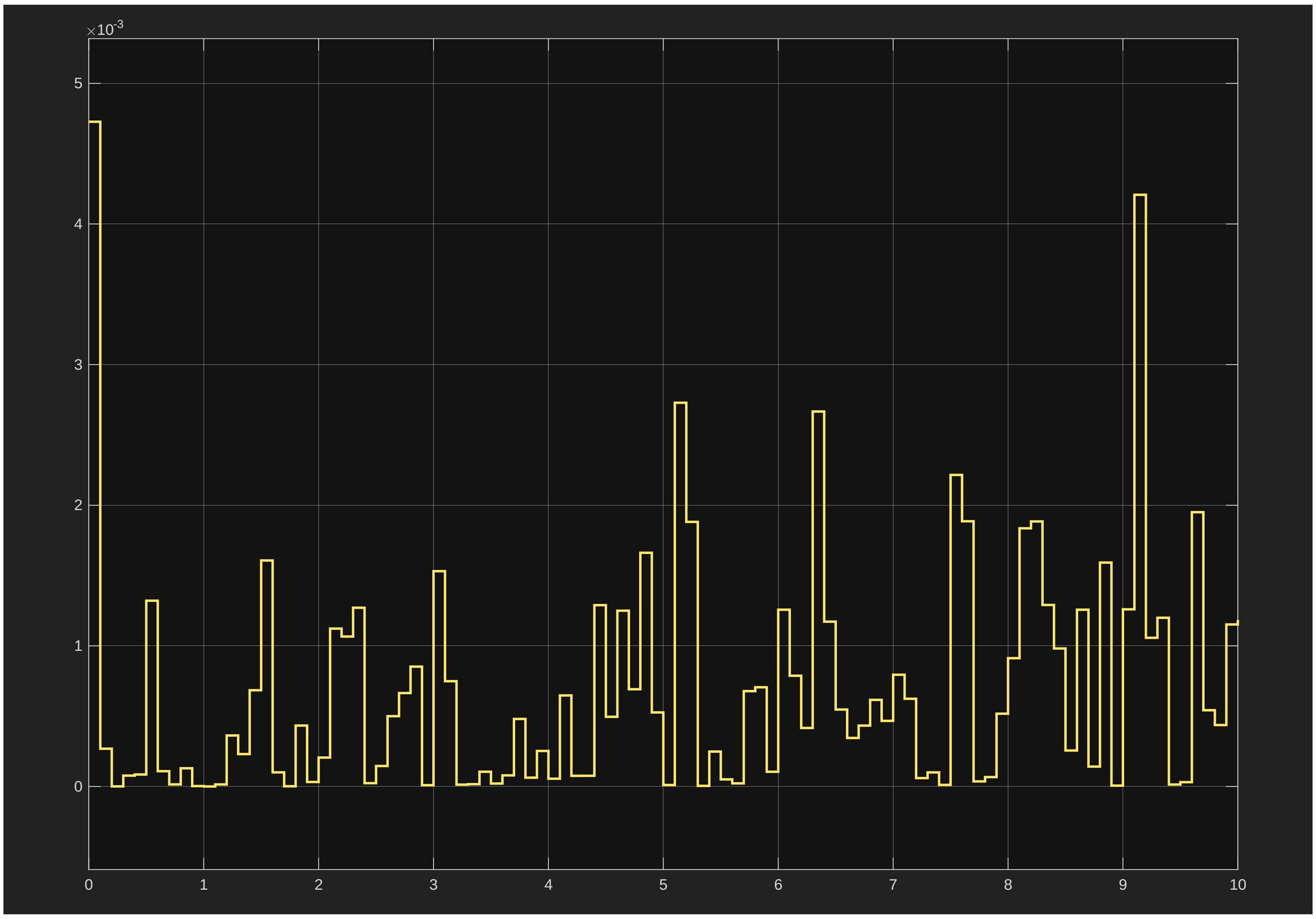

Figure 4: Mean Square Error of the ideal OTA computation network over an AWGN channel.

rate (`sps_actual = sps_design = 6`). In other words the upsampling rate of the FIR filters matches the spacing between sample generation.

An important architectural consideration emerges from the GNU Radio implementation: in the absence of explicit digital modulation schemes, strategic downsampling at the transmitter becomes mandatory to enable proper filter operation. Specifically, each individual sensor transmits one symbol while simultaneously discarding sixteen interpolated samples, thereby maintaining compliance with Nyquist sampling criterion while leveraging the spectral efficiency benefits of pulse shaping.

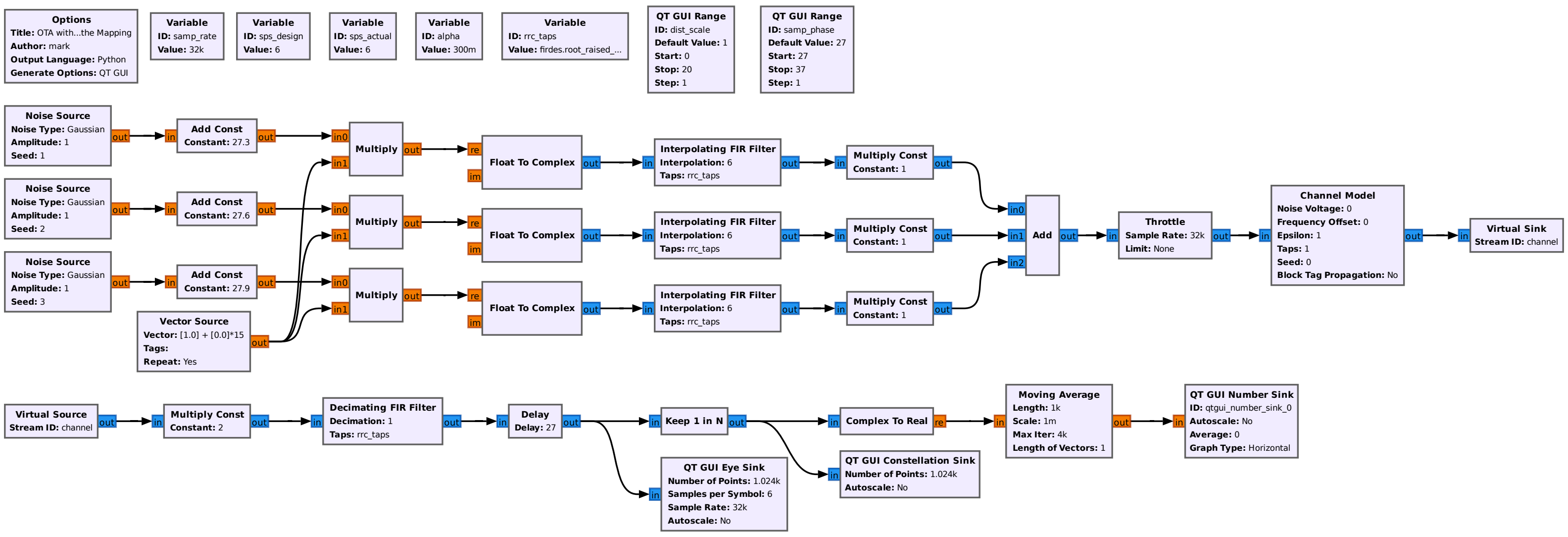

Figure 5: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

Under these constraints, the Eye Diagram remains perfectly ”open” at the optimal sampling phase, and the Constellation Diagram exhibits tightly clustered points. The absence of horizontal dispersion in the constellation confirms that, prior to the introduction of timing offsets or Faster-Than-Nyquist signaling, the MAC superposition principle flawlessly computes the arithmetic average of the agricultural sensors.

A critical implementation detail warranting emphasis concerns the optimal sampling phase selection. Empirical observation from the GNU Radio receiver indicates that maximum data recovery is achieved at a sampling phase offset of 33 samples out of the 96-sample cycle ([1]). This phase offset reflects the group delay inherent to the polyphase interpolating and decimating FIR filter implementations, necessitating careful calibration for optimal demodulation performance.

[1]The fundamental cycle spans 16 symbols; with 6 samples per symbol, the total cycle length is $16 \times 6 = 96$ samples.

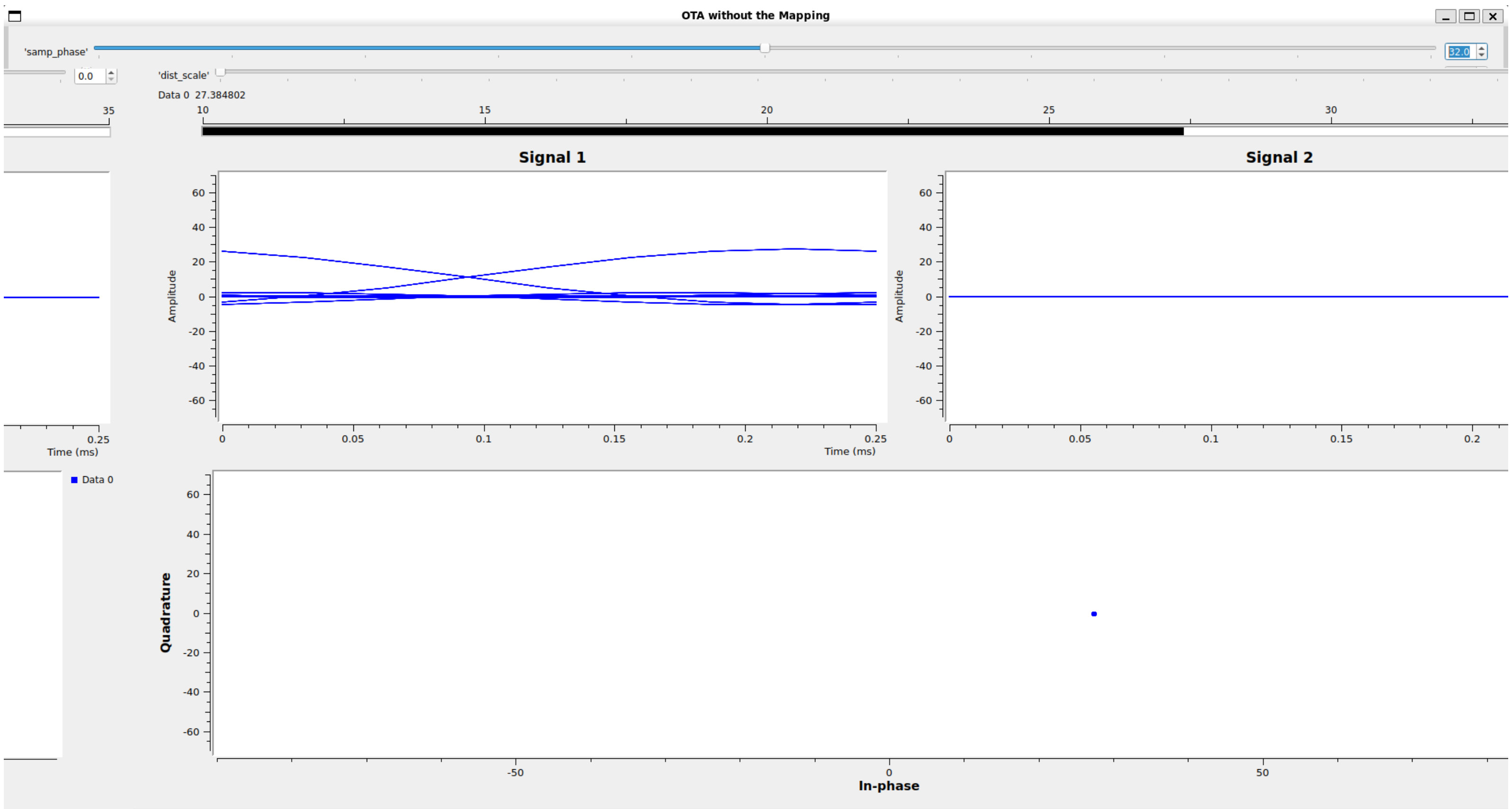

Figure 6: GNU Radio Eye & Constelation Diagram for the ideal, Nyquist-compliant scenario.

References

[1] X. Cao, G. Zhu, J. Xu, Z. Wang, and S. Cui, "Optimized power control design for over-the-air federated edge learning," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 1, pp. 342–358, Jan. 2022, ISSN: 0733-8716, 1558-0008. DOI: 10.1109/JSAC.2021.3126060 Accessed: Dec. 4, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/9606731/

[2] M. Goldenbaum and S. Stanczak, "Robust analog function computation via wireless multiple-access channels," *IEEE Transactions on Communications*, vol. 61, no. 9, pp. 3863–3877, Sep. 2013, ISSN: 0090-6778. DOI: 10.1109/TCOMM.2013.072913.120815 Accessed: Dec. 4, 2025. [Online]. Available: http://ieeexplore.ieee.org/document/6573232/

[3] A. Şahin and R. Yang, "A survey on over-the-air computation," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 3, pp. 1877–1908, 2023, ISSN: 1553-877X, 2373-745X. DOI: 10.1109/COMST.2023.3264649 Accessed: Dec. 4, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10092857/

[4] G. Zhu, Y. Du, D. Gunduz, and K. Huang, "One-bit over-the-air aggregation for communication-efficient federated edge learning: Design and convergence analysis," *IEEE Transactions on Wireless Communications*, vol. 20, no. 3, pp. 2120–2135, Mar. 2021, ISSN: 1536-1276, 1558-2248. DOI: 10.1109/TWC.2020.3039309 Accessed: Dec. 4, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/9272666/